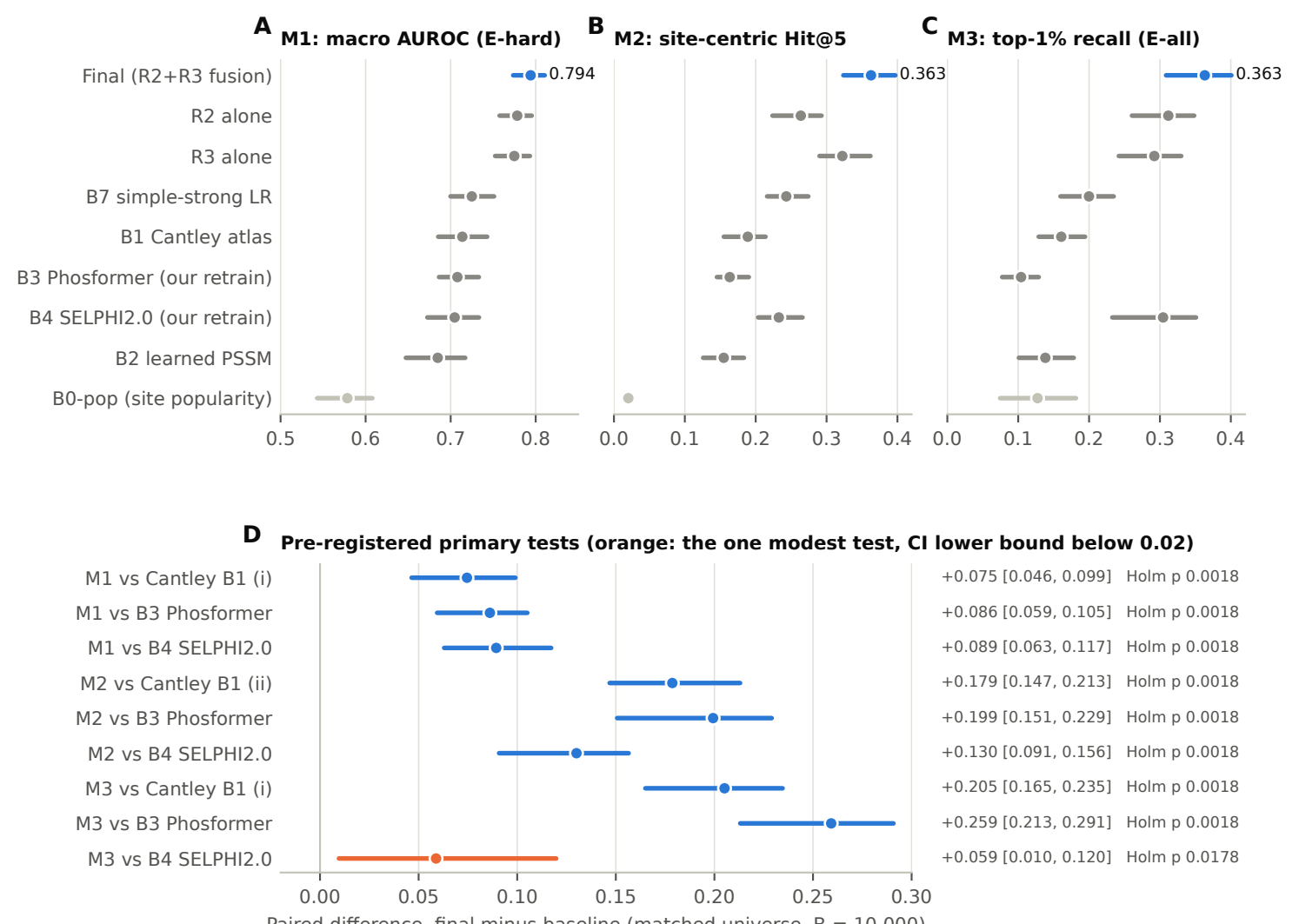
A
M1: macro AUROC (E-hard)
B
M2: site-centric Hit@5
C
M3: top-1% recall (E-all)
Final (R2+R3 fusion)
R2 alone
R3 alone
B7 simple-strong LR
B1 Cantley atlas
B3 Phosformer (our retrain)
B4 SELPHI2.0 (our retrain)
B2 learned PSSM
B0-pop (site popularity)
0.794
0.363
0.363
0.5
0.6
0.7
0.8
0.0
0.1
0.2
0.3
0.4
0.0
0.1
0.2
0.3
0.4
D
Pre-registered primary tests (orange: the one modest test, CI lower bound below 0.02)
M1 vs Cantley B1 (i)
M1 vs B3 Phosformer
M1 vs B4 SELPHI2.0
M2 vs Cantley B1 (ii)
M2 vs B3 Phosformer
M2 vs B4 SELPHI2.0
M3 vs Cantley B1 (i)
M3 vs B3 Phosformer
M3 vs B4 SELPHI2.0
+0.075 [0.046, 0.099] Holm p 0.0018
+0.086 [0.059, 0.105] Holm p 0.0018
+0.089 [0.063, 0.117] Holm p 0.0018
+0.179 [0.147, 0.213] Holm p 0.0018
+0.199 [0.151, 0.229] Holm p 0.0018
+0.130 [0.091, 0.156] Holm p 0.0018
+0.205 [0.165, 0.235] Holm p 0.0018
+0.259 [0.213, 0.291] Holm p 0.0018
+0.059 [0.010, 0.120] Holm p 0.0178
0.00
0.05
0.10
0.15
0.20
0.25
0.30
Paired difference, final minus baseline (matched universe, B = 10,000)